

Swan for the victims, but certainly not to the perpetrators. The Postscript provides an additional discussion of the point.

- * The Idea of Robustness: Why do we formulate theories leading to projections and forecasts without focusing on the robustness of these theories and the consequences of the errors? It is much easier to deal with the Black Swan problem if we focus on robustness to errors rather than improving predictions.
- * *Recursive* here means that the world in which we live has an increasing number of feedback loops, causing events to be the cause of more events (say, people buy a book *because* other people bought it), thus generating snowballs and arbitrary and unpredictable planet-wide winner-take-all effects. We live in an environment where information flows too rapidly, accelerating such epidemics. Likewise, events can happen *because* they are not supposed to happen. (Our intuitions are made for an environment with simpler causes and effects and slowly moving information.) This type of randomness did not prevail during the Pleistocene, as socioeconomic life was far simpler then.
- * The metaphor of the black swan is not at all a modern one—contrary to its usual attribution to Popper, Mill, Hume, and others. I selected it because it corresponds to the ancient idea of a “rare bird.” The Latin poet Juvenal refers to a “bird as rare as the black swan”—*rara avis in terris nigroque simillima cygno*.
- * It is also naïve empiricism to provide, in support of some argument, series of eloquent confirmatory quotes by dead authorities. By searching, you can always find someone who made a well-sounding statement that confirms your point of view—and, on every topic, it is possible to find another dead thinker who said the exact opposite. Almost all of my non-Yogi Berra quotes are from people I disagree with.